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Describe how a triglyceride molecule is formed.

[3 marks]

0 8 . 2

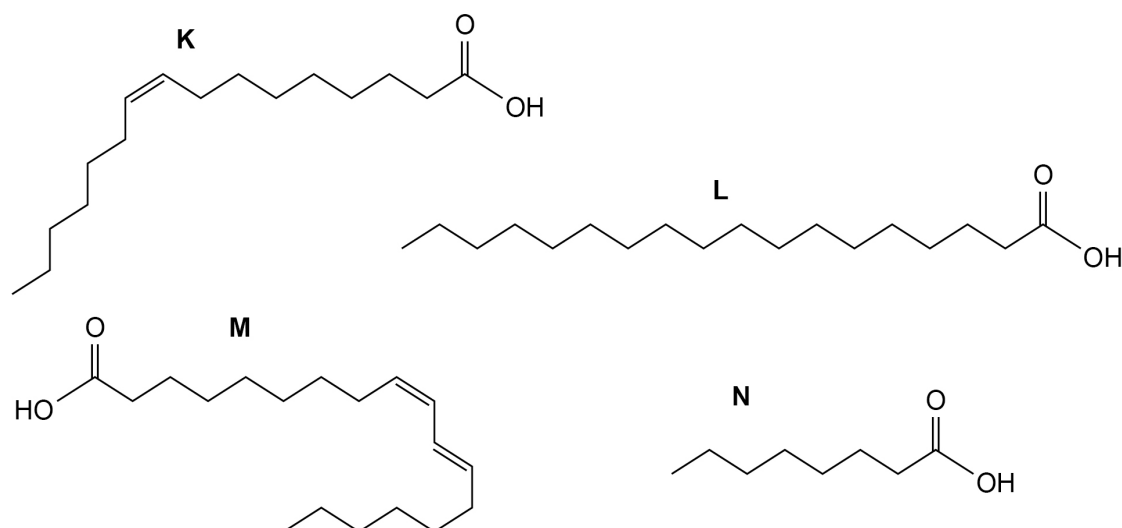
Table 5 shows some properties of four fatty acids.

Table 5

Fatty acid	Number of carbon atoms in the R group	Number of double bonds in the R group
Caprylic acid	8	0
Palmitoleic acid	16	1
Stearic acid	18	0
Linoleic acid	18	2

Figure 10 shows diagrams of these fatty acids.

Figure 10

Put a tick (✓) in **one** box that contains correct information about one of these fatty acids.

[1 mark]

Caprylic acid is an unsaturated fatty acid represented by diagram L.

☐

Linoleic acid is a saturated fatty acid represented by diagram N.

☐

Palmitoleic acid is an unsaturated fatty acid represented by diagram K.

☐

Stearic acid is a saturated fatty acid represented by diagram M.

☐

Turn over ►

The percentage of saturated fatty acids compared with unsaturated fatty acids found in lipid stores in seeds differs in different populations.

Scientists investigated two populations of the plant, *Helianthus annuus*.

The scientists grew young plants from seeds collected from each population. They placed the seeds on wet tissue paper so that the root growth was visible.

They grew seeds from each population at two temperatures:

- warm temperature of 24 °C
- cool temperature of 10 °C

After 10 days, the scientists measured the length of each root.

Table 6 shows some of the properties of the two populations and the scientists' results.

Table 6

Population	Temperature in natural environment	In the seed – Mean percentage of fatty acids that are saturated	Mean length of root after 10 days at 24 °C / mm ($\pm 2 \times$ standard deviation)	Mean length of root after 10 days at 10 °C / mm ($\pm 2 \times$ standard deviation)
1	Warm	10.9	8.2 (± 1.0)	3.1 (± 0.3)
2	Cool	6.1	5.5 (± 0.9)	4.3 (± 0.2)

The mean $\pm 2 \times$ standard deviation includes 95% of the data.

0 8 . 3

The scientists used a data logger to measure the length of the root rather than a ruler.

Suggest **one** reason why they used a data logger **and** explain why this was important in this investigation.

[1 mark]

0 8 . 4

It is known that:

- during respiration saturated fatty acids yield more energy than unsaturated fatty acids
- saturated fatty acids have higher melting points than unsaturated fatty acids
- lipases in seeds act more rapidly on liquid substrates.

Use this information and **Table 6** to show how each population is better adapted for its natural environment when compared with the other population.

[4 marks]

0 8 . 5

Although these two populations are completely separate and show genetic variation, they are both called *Helianthus annuus*.

Explain why they are both given this name.

[1 mark]
